

Development of an Artificial Intelligence Algorithm for Forest Fire Risk Assessment



SPACE CHALLENGES

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August 18, 2025



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1 Introduction

Forest fires have become an increasingly frequent and devastating phenomenon in recent decades, driven primarily by the escalating impacts of anthropogenic climate change. Rising global temperatures, prolonged droughts, and shifts in vegetation patterns have contributed to more intense and widespread fire seasons around the world. This growing threat poses serious challenges not only to forest ecosystems, but also to human settlements, critical infrastructure, and public health systems. The first half of 2025 saw intense global wildfire activity, with events in Canada, Greece, Chile, Australia, the USA, and Indonesia resulting in billions in insured losses, burned areas spanning hundreds of thousands of hectares, and other significant consequences such as mass evacuations, severe air pollution, and the destruction of homes:

Country	Wildfire Event (Date)	Insured Losses (€)	Burned Area (hectares)	Other Consequences
Canada	Alberta wildfires (May 2025)	1.2 billion	1,500,000	Mass evacuations, severe air pollution
Greece	Rhodes wildfires (July 2025)	500 million	25,000	Tourism disruption, biodiversity loss
Chile	Valparaíso fires (Feb 2025)	300 million	50,000	Hundreds of homes destroyed
Australia	Victoria bushfires (Jan 2025)	800 million	600,000	Wildlife devastation, long-term drought
USA	California fires (June 2025)	2.5 billion	200,000	Power outages, insurance crises
Indonesia	Sumatra peat fires (Mar 2025)	150 million	100,000	Transboundary haze, health emergencies

Figure 1: The forest fire consequences

These events underscored the limitations of traditional fire management methods and highlighted the critical and urgent need for more advanced, data-driven predictive approaches. Such approaches could operate at scale and provide targeted, actionable intelligence in time for risk mitigation and response activation measures to be applied.



Figure 2: Recent wildfire event in California, United States of America (USA), demonstrating the scale and intensity of modern forest fires

Forest wildfire prevention and management require accurate, localized and timely risk assessments. Traditional fire management methods often lacked the comprehensive spatial coverage and high-frequency data needed to address the scale of modern wildfire challenges. A key solution to this problem is the creation of



predictive systems that can analyze environmental variables with high frequency and precise geolocalization, to allow more efficient forest wildfire prevention and management.

Recent advancements in remote sensing technologies and machine learning have started providing new opportunities for high-accuracy fire risk assessment. The revolutionary transformation of the space sector, commonly referred to as the **New Space** era [1, 2], characterized by reduced launch cost and miniaturization of the technology. This paradigm shift has fundamentally democratized access to space-based capabilities [3]. No longer limited to major government agencies such as National Aeronautics and Space Administration (NASA), European Space Agency (ESA), or other national space organizations, space-based Earth observation is now accessible to universities, research institutions, startups, and small private companies. This democratization has enabled the development of innovative applications and services on a scale that was previously technically and economically unimaginable.

One of the most impactful applications that emerged from this New Space revolution was comprehensive Earth observations for environmental monitoring and disaster risk assessment. With the deployment of constellations of small, cost-effective satellites equipped with advanced optical, thermal, and hyperspectral sensors, it is now feasible to monitor key environmental variables with high temporal resolution (daily or sub-daily revisit times) and broad spatial coverage [4]. This allowed for the collection of crucial environmental data like vegetation health indices (Normalized Difference Vegetation Index (NDVI)), land surface temperature (Land Surface Temperature (LST)), soil moisture content etc. on a daily or sub-daily basis, and with continuously increasing spatial resolution.



Figure 3: The Frame SmallSat platform from EnduroSat, exemplifying the New Space approach to affordable Earth observation

By integrating this rich, multi-spectral satellite imagery with deep learning methodologies, we aimed to create a robust, scalable system capable of providing accurate fire risk predictions across diverse biomes with vegetation and environmental conditions. This research has contributed to the growing body of knowledge at the intersection of remote sensing, machine learning, and environmental disaster management, while demonstrating the practical potential of New Space technologies for addressing critical societal challenges.



1.1 Contribution

This research made several key contributions to the field of forest fire risk assessment and satellite-based Earth observation:

Methodological Contributions: We integrated multi-spectral Sentinel-2 satellite imagery with deep learning architectures tailored for forest fire risk assessment. This hybrid approach leverages neural networks to capture both spatial patterns and temporal dynamics, which are essential for accurate risk prediction.

Technical Innovation: We developed a modular system architecture that supports scalable deployment across diverse geographical regions while preserving high prediction accuracy. The cloud detection and forest classification modules achieved state-of-the-art results on standard benchmarks, providing robust preprocessing capabilities for real-world and even on-board operational use.

Broader Implications: This work highlights the transformative potential of New Space technologies for tackling critical environmental challenges. The proposed framework can be adapted to related applications such as drought monitoring, ecosystem health assessment, and climate change impact analysis.

1.2 Document Structure

This document is organized into nine comprehensive sections that systematically present the development, implementation, and evaluation of our AI-driven forest fire risk assessment framework:

Section 2 (Related Work and State of the Art) provides a critical review of existing research in forest fire risk prediction, satellite-based environmental monitoring, and machine learning applications in Earth observation, identifying key gaps and positioning our contribution within the current scientific landscape.

Section 3 (Requirements) establishes the functional and performance requirements for the proposed system, including technical specifications, accuracy targets, and operational constraints derived from real-world fire management needs.

Section 4 (Dataset) describes the comprehensive multi-source datasets employed in this research, including cloud detection (Section 4.1), vegetation detection (Section 4.2) and fire prediction (Section 4.3) datasets, along with data preprocessing and validation methodologies.

Section 5 (System Architecture) presents the detailed technical architecture of our AI framework, encompassing label creation and feature extraction techniques (Section 5.1 and 5.4), Cloud detection algorithms (Section 5.2), (Section 5.3), and the core fire risk assessment model (Section 5.5).

Section 6 (Experimental Results and Evaluation) provides comprehensive experimental validation of the proposed system, including quantitative performance metrics, comparative analysis with existing approaches,



and assessment across different environmental conditions and forest types.

Section 7 (Discussion and Limitations) offers critical analysis of the results, identifies current limitations of the approach, and discusses potential sources of uncertainty and areas for future improvement.

Section 8 (Business and Societal Impact) examines the broader implications of this research, including potential applications in operational fire management, economic benefits, and contributions to environmental protection and public safety.

Section 9 (Team and Work Breakdown) summarizes the team work and efforts done by each team member.

Section 10 (Conclusions) summarizes the key findings, contributions, and future research directions emerging from this work.

2 Related Work and State of the Art

The intersection of remote sensing, machine learning, and forest fire risk assessment is a rapidly evolving research domain that has gained significant momentum over the past two decades. This section provides a comprehensive review of existing approaches, categorizing them into distinct research streams while critically evaluating their contributions, limitations, and relevance to the proposed methodology.

2.1 Traditional Remote Sensing Approaches for Fire Risk Assessment

Early approaches to satellite-based fire risk assessment primarily relied on conventional image processing techniques and statistical methods applied to optical and thermal remote sensing data. These foundational works established many of the fundamental principles that continue to inform current research.

Traditional vegetation indices, particularly the NDVI [5], have served as cornerstone indicators for assessing vegetation stress and fire susceptibility. While computationally efficient and interpretable, these approaches often exhibit limited sensitivity to subtle environmental changes, such as minor decreases in fuel moisture content, which can be critical for early fire risk prediction. For example, a common shortcoming is that NDVI-based models may fail to predict fire risk in regions with high elevation, low moisture vegetation that does not show a clear change in greenness, and struggle to capture the complex, non-linear relationships between multiple environmental variables [6]. Additionally, traditional methods typically rely on static thresholds and rule-based decision systems that may not generalize well across different geographical regions and climatic conditions.

Thermal remote sensing approaches have focused primarily on detecting active fires and burn scars rather than predicting fire risk [7]. While valuable for post-event analysis and immediate fire detection, these methods



provide limited predictive capability for risk assessment applications. The temporal resolution constraints of traditional satellite missions also limit their effectiveness for real-time monitoring applications.

2.2 Machine Learning Applications in Fire Risk Prediction

The integration of machine learning techniques with remote sensing data has marked a significant paradigm shift in fire risk assessment research. Jain et al. [8] provide a comprehensive survey of machine learning applications in wildfire science, highlighting both the potential and current limitations of these approaches.

Early machine learning applications employed classical algorithms such as decision trees, random forests, and support vector machines applied to hand-crafted features derived from satellite imagery. While these approaches demonstrated improved predictive performance compared to traditional statistical methods, they remained constrained by the quality and representativeness of manually engineered features. For instance, a common function shortcoming is that a Random Forest model trained on simple spectral indices and meteorological data may achieve F1 - score of 0.75 in a specific Californian study area, but its performance degrades significantly when applied to the dry forest of Australia without extensive feature re - engineering. Furthermore, most of these studies were limited to specific geographical regions and failed to demonstrate transferability across different environmental conditions.

More recent work has explored ensemble methods and advanced feature selection techniques to improve model robustness and generalization. However, the majority of existing machine learning approaches still rely on relatively simple feature representations and struggle to capture the temporal dynamics and spatial patterns that are crucial for accurate fire risk assessment.

2.3 Deep Learning and Advanced AI Techniques

The emergence of deep learning methodologies has opened new possibilities for automated feature extraction and pattern recognition in satellite imagery analysis. Zhu et al. [9] provide a comprehensive review of deep learning applications in remote sensing, highlighting the potential for Convolutional Neural Networks (CNNs) and other deep architectures to automatically learn relevant features from raw satellite data.

However, the application of deep learning to fire risk assessment specifically remains limited. Most existing deep learning studies in this domain focus on post-fire damage assessment or active fire detection rather than predictive risk modeling. For example, a CNN model might be highly accurate at classifying burned areas after a fire, but it may not be architecturally designed to ingest time-series data to predict the likelihood of a fire starting in the future. Furthermore, the computational requirements and data demands of deep learning approaches often limit their applicability to operational fire management scenarios, particularly in resource-constrained environments.



Current deep learning approaches also face challenges in handling the multi-temporal, multi-spectral nature of satellite data required for comprehensive fire risk assessment. While some studies have explored recurrent neural networks (Recurrent Neural Networks (RNNs)) and long short-term memory (Long Short-Term Memory (LSTM)) networks for temporal modeling, these approaches have not been systematically validated across diverse environmental conditions and temporal scales.

For image-to-image regression tasks like fire prediction, one of the state-of-the-art architectures in the Deep Learning field is the **U-Net** [10], [11]. This architecture is a derivative network of a *convolutional autoencoder*.

2.3.1 Introduction to Convolutional Autoencoders

Convolutional autoencoders represent a specialized class of neural networks that combine the spatial feature extraction capabilities of convolutional neural networks with the dimensionality reduction and reconstruction principles of traditional autoencoders. These architectures consist of two primary components: an encoder that progressively reduces spatial dimensions while extracting hierarchical features, and a decoder that reconstructs the input by upsampling the encoded representation back to the original resolution. The encoder utilizes convolutional layers with pooling operations to create increasingly abstract feature representations, while the decoder employs transposed convolutions or upsampling layers to restore spatial resolution. This encoder-decoder structure makes convolutional autoencoders particularly suitable for tasks requiring pixel-level predictions, as they can learn to map input images to output images of the same dimensions while capturing essential spatial relationships and patterns within the data.

The application of convolutional autoencoders to a task of forest fire prediction or cloud masking tasks stems from their inherent ability to perform pixel-wise predictions, which in these cases are continuous values between 0 and 1, representing the cloud coverage, fire probability or risk levels for each spatial location. Forest fire prediction fundamentally requires the analysis of complex spatial patterns, environmental conditions, and temporal dynamics that are naturally embedded within satellite imagery and remote sensing data. Convolutional autoencoders excel in this domain because they can learn to identify subtle visual indicators of fire risk, such as vegetation density, moisture content, topographical features, and meteorological conditions, all of which contribute to fire susceptibility. Furthermore, the spatial coherence maintained by convolutional operations ensures that neighboring pixels with similar environmental characteristics receive consistent risk assessments, and that the neural network performs *contextual prediction*, rather than basing its outputs only on single-element values.

2.3.2 U-Net Architecture and Its Advantages

U-Net represents a significant advancement over traditional convolutional autoencoders through its innovative incorporation of *skip connections* that directly link corresponding layers in the encoder and decoder pathways.



These connections were originally introduced in the *ResNet* architecture [12], to avoid the *vanishing gradient* problem, which can occur in very deep networks, causing earlier layers to receive too small updates. ResNet solved this by letting the gradient flow directly from the output to the input of the network, "skipping" the intermediate layers. Named after its distinctive U-shaped architecture, U-Net was originally developed for biomedical image segmentation but has since become a fundamental architecture for various pixel-level prediction tasks. In the U-Net, the skip connections concatenate high-resolution feature maps from the encoder

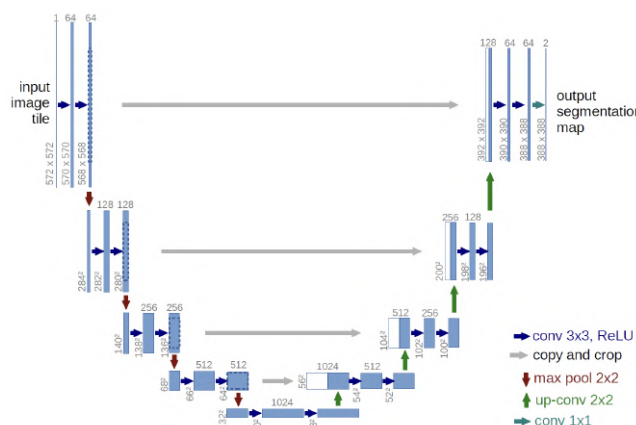


Figure 4: The original U-Net architecture. The skip-connections between corresponding encoder and decoder layers can be seen, indicated as "copy and crop". The last two-channel layer are used for the creation of the binary mask for the image segmentation task in [10]

directly to the corresponding decompressed features in the decoder layers, effectively bypassing the information bottleneck that occurs at the deepest layer of traditional autoencoders, and allowing to combine the high-level features extracted by the encoder with decompressed features of the decoder.

All these characteristics make the U-Net a highly effective architecture for image2image tasks like semantic segmentation, or pixel-wise prediction - including forest fire prediction and cloud detection applied to remotely sensed images.

2.4 Satellite Technology and New Space Applications

The New Space revolution has fundamentally transformed the landscape of Earth observation capabilities available for environmental monitoring applications [1, 2]. The proliferation of small satellite constellations has enabled unprecedented temporal resolution and global coverage, creating new opportunities for real-time environmental monitoring[3]. This has enabled the creation of vast datasets that contain imagery from Earth observation, which could be found to be particularly useful for training A.I. models.

Modern satellite missions, particularly the European Space Agency's Copernicus Sentinel constellation [13], provide systematic, high-resolution multi-spectral imagery with regular revisit cycles that are well-suited for fire risk monitoring applications. The combination of multiple spectral bands, including near-infrared and shortwave



infrared channels, enables more sophisticated analysis of vegetation health and moisture content compared to traditional optical imagery.

However, the integration of New Space technologies with operational fire management systems remains limited. Most current applications focus on research contexts rather than operational deployment, and there is a significant gap between the technical capabilities of modern satellite systems and their practical implementation in real-world fire management workflows. A specific functional shortcoming of this gap is that new, high-resolution imagery from commercial satellite providers may use proprietary data formats that are incompatible with the legacy software used by government fire management agencies.

2.5 Positioning of Our Contribution

Our proposed approach addresses these identified gaps by developing a comprehensive AI framework that integrated state-of-the-art deep learning methodologies with multi-spectral Sentinel-2 satellite imagery for real-time forest fire risk assessment. Key differentiators of our approach included:

Multi-Component Architecture: Unlike existing approaches that focus on single aspects of the problem, our framework integrated cloud detection, vegetation detection, feature extraction, and risk assessment into a unified system architecture.

Temporal Dynamics Modeling: Our approach explicitly modeled the temporal evolution of environmental conditions through advanced sequence modeling techniques, enabling more accurate risk prediction compared to static approaches.

Scalable Implementation: The framework was designed with operational deployment considerations, including computational efficiency and real-time processing capabilities suitable for New Space satellite constellations.

This positioned contribution represented a significant advancement in the state of the art, bridging the gap between cutting-edge AI techniques and practical fire management applications while leveraging the full potential of modern satellite technology infrastructure.

3 Requirements

In this section we are going to discuss the different requirements that our product should have.



3.1 Functional Requirements

Req. ID	Description	Priority
FR-01	The system shall process multi-spectral Sentinel-2 satellite imagery with all relevant spectral bands	High
FR-02	The system shall handle cloud detection and masking with minimum 85% accuracy	High
FR-03	The system shall detect vegetation areas from satellite imagery	High
FR-04	The system shall extract relevant environmental features including vegetation indices and soil moisture estimates	High
FR-05	The system shall generate fire risk predictions with spatial resolution compatible with Sentinel-2 data (10-20m pixel size)	High
FR-06	The system shall generate risk heatmaps	Medium
FR-07	The system shall maintain prediction accuracy of at least 90% when validated against historical fire occurrence data	High
FR-8	The system shall provide confidence intervals or uncertainty estimates for risk predictions	Medium
FR-9	The system shall support real-time or near real-time risk assessment updates	Medium

3.2 Non-Functional Requirements

Req. ID	Description	Priority
NFR-01	The system shall scale to process multiple scenes concurrently for large-area assessments	Medium
NFR-02	The system shall maintain consistent processing times regardless of geographical region or season	Medium
NFR-03	The system shall demonstrate robust performance across different forest biomes and climatic conditions	High
NFR-04	The system shall maintain data integrity throughout all processing stages	High
NFR-05	The system shall provide intuitive interfaces for fire management personnel	Medium
NFR-06	The system shall include comprehensive documentation and user guides	Low
NFR-07	The system shall provide clear error messages and diagnostic information	Low



3.3 Technical Requirements

Req. ID	Description	Priority
TR-01	The system shall be deployed on an NVIDIA Jetson	High
TR-02	Python 3.11+ runtime environment with docker deployment	High
TR-03	Deep learning framework support (PyTorch)	High
TR-04	Geospatial data processing capabilities (GDAL, rasterio...)	High
TR-05	Historical fire occurrence data for model training and validation	High

3.4 Regulatory and Compliance Requirements

Req. ID	Description	Priority
RCR-01	The system shall comply with relevant data protection and privacy regulations	High
RCR-02	The system shall meet standards for scientific reproducibility and transparency	High
RCR-03	The system shall provide audit trails for all processing decisions and model predictions	Medium
RCR-04	The system shall support verification and validation procedures required for operational use	High

4 Dataset

Every model needs a solid foundation on which to be built, and in our case, that foundation is data. The selection, preparation, and understanding of the datasets not only define the scope of the model but also unlock its potential. This section describes the data sources used, their most relevant characteristics, and the criteria followed to ensure their quality and suitability for the problem at hand.

Since our project involves different approaches and models, it was necessary to work with significant volumes of data, meeting high standards of accuracy and specificity. Each dataset offers a distinct perspective and plays a key role within the experimental pipeline. Below, we detail their particularities and how they were integrated into the various stages of development.

4.1 Cloud Detection Dataset

Accurate cloud detection constitutes a fundamental preprocessing requirement for satellite-based Earth observation applications, as cloud contamination can significantly compromise the quality and reliability of surface

reflectance measurements essential for vegetation monitoring and fire risk assessment. The presence of clouds introduced systematic errors in spectral indices, obscured critical surface features, and created temporal gaps in monitoring capabilities that can coincide with critical fire weather periods.

4.1.1 Dataset Selection and Characteristics

For the development and validation of our cloud detection algorithms, we primarily utilized **38-Cloud dataset** for training and the **CloudSEN12 dataset** for validation [14], both datasets represent a comprehensive benchmark dataset specifically designed for semantic segmentation of Sentinel-2 satellite imagery. The overall datasets comprise more than 10000 Sentinel-2 Level-1C subscenes.



Figure 5: Representative example from CloudSEN12 dataset showing Sentinel-2 imagery (left) with corresponding ground truth segmentation mask (right).

4.2 Vegetation Detection Dataset

4.2.1 Requirements

A dataset with accurately labeled satellite imagery of vegetation and non-vegetation was needed for the proper testing of the vegetation detector algorithm. The dataset needed to have at least the Red and Near-Infrared (NIR) bands. With these bands, the Normalized Difference Vegetation Index (NDVI) was calculated:

$$NDVI = \frac{NIR - RED}{NIR + RED} \quad (1)$$



4.2.2 Datasets Used

The dataset used was taken from the official Copernicus website. The dataset was manually split into two labels of .NPY files: those with vegetation and those without vegetation on the land. Data augmentation was applied to the dataset by applying a set of geometric transformations: a counter-clockwise 90° rotation, 180° rotation, a mirrored horizontal flip, and a mirrored vertical flip. The files were taken from a focused image of Chile and augmented to 256 by 256 pixels. At the end 200 .NPY files were extracted for testing the vegetation detection model.

4.2.3 Known Limitations

The Copernicus website provides useful data for testing the Vegetation Detection Model. However, the user should be aware of several limitations that the dataset consists of:

1. Geographic bias - Most of the Sentinel-2 files are Chile focused. This should not diminish the credibility of the dataset as it is only used for testing, but should be taken into account as it means that the model has not been tested on all available biomes on Earth.
2. Limited classes – labeling the dataset into “vegetation” and “no vegetation” will make the model indiscriminate to different types of vegetations (e.g., the model would not distinguish between coniferous and deciduous forest types).

4.3 Fire Risk Assessment Dataset

Understanding and anticipating wildfire events requires accurate, temporally-aligned observations of affected regions both before and after the fire. To support the development of models for fire risk assessment, we curated a dataset composed of paired satellite images—capturing the same location prior to ignition and after. These image pairs provide vital insights into the visual and spectral changes that occur due to fire, which are key for training and validating machine learning algorithms aimed at fire detection and risk evaluation.

Our dataset construction began by identifying past real-world wildfire events reported in reputable sources, with a focus on the European Forest Fire Information System (EFFIS) official website of Copernicus. These reports provided detailed information on the geographic location and the dates of known forest fire outbreaks across the globe. Based on this data, we selected specific regions and timeframes to extract satellite imagery from the open-access Copernicus Sentinel-2 dataset. To ensure relevance and consistency, two images were retrieved for each event: one representing a pre-fire (safe) state, and another showing the after-fire (post-ignition) state of the same area.



To characterize vegetation and burned areas, we relied on specific Sentinel-2 spectral bands—Band 8 (near-infrared) and Band 12 (shortwave infrared). These bands were selected due to their sensitivity to vegetation health and moisture content:

- Band 8 (NIR): Captures reflectance from healthy vegetation, which strongly reflects near-infrared light, due to the presence of spongy mesophylls.
- Band 12 (SWIR): This band is sensitive to vegetation moisture and thermal anomalies, making it particularly useful in identifying burned or stressed vegetation.

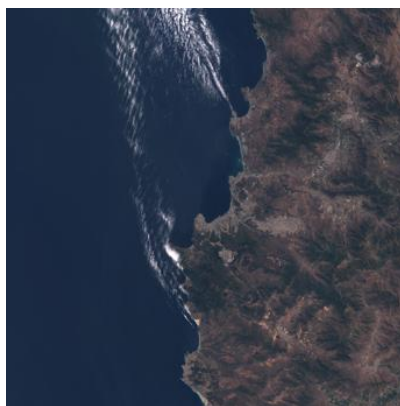
Specifically, using these bands, the *Normalized Burn Ratio* (NBR) was calculated:

$$NBR = \frac{B08 - B12}{B08 + B12} \quad (2)$$

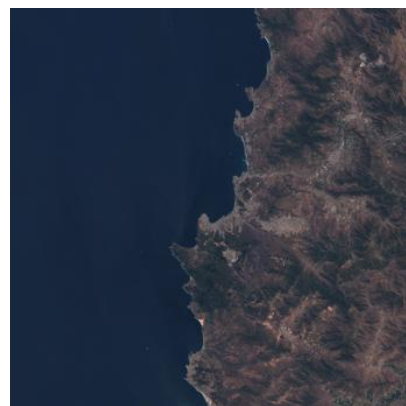
This index enables us to quantify the difference in vegetation condition before and after fire events. Values of the index range from -1 to 1:

- High positive values typically correspond to healthy, unburned vegetation.
- Lower or negative values indicate burned areas, bare soil, or sparse vegetation.

By analyzing this index across the temporal image pairs, we can clearly observe significant decreases in index values in the post-fire images. This change reliably highlights burned regions and forms the basis for both supervised learning and threshold-based classification approaches. An example of the dataset can be seen from the true colour visualization (Figure 6) of Valparaíso, Chile, before and after a forest wildfire on February 7, 2024, and the NBR index visualization (Figure 7), where it can be better appreciated the change in appearance of the image.



(a) Before wildfire



(b) After wildfire

Figure 6: Valparaíso, Chile before and after the forest wildfire on February 7, 2024 (without NBR)

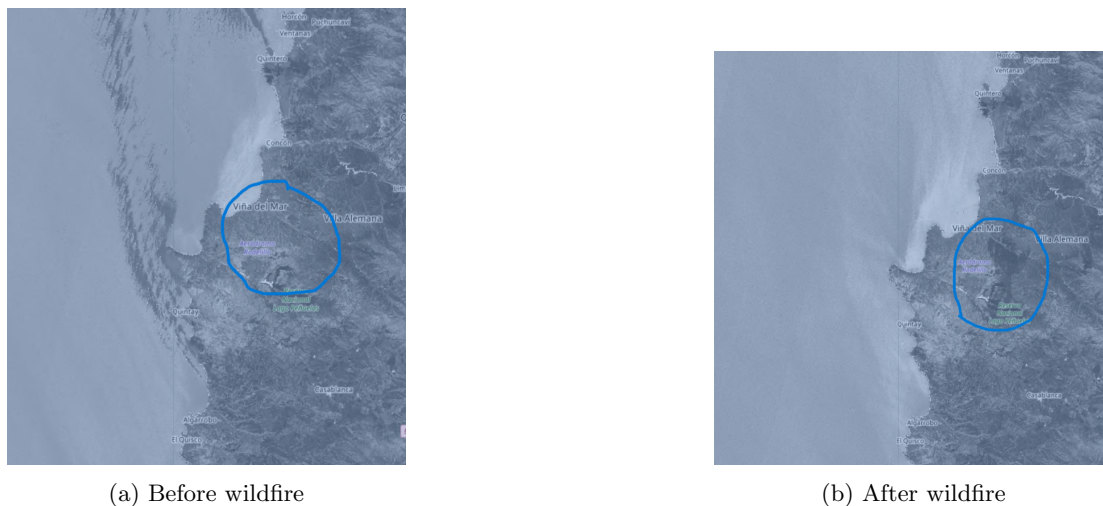


Figure 7: Valparaíso, Chile NBR index before (a) and after (b) the forest wildfire on 07.02.2024

The inclusion of this dataset into our experimental pipeline provides an essential perspective for assessing fire-related risks. Its alignment with real-world events, spectral robustness, and geospatial accuracy ensures that our AI models are trained on relevant, high-impact scenarios—enabling more accurate detection and risk forecasting in future deployments.

5 System Architecture

Our Artificial Intelligence (AI)-driven forest fire risk assessment system employs a modular architecture designed to process multi-spectral Sentinel-2 satellite imagery through a sequential pipeline of specialized components. This section presents the detailed technical design and implementation of each system component, emphasizing the integration of state-of-the-art deep learning methodologies with operational requirements for real-time fire risk assessment.

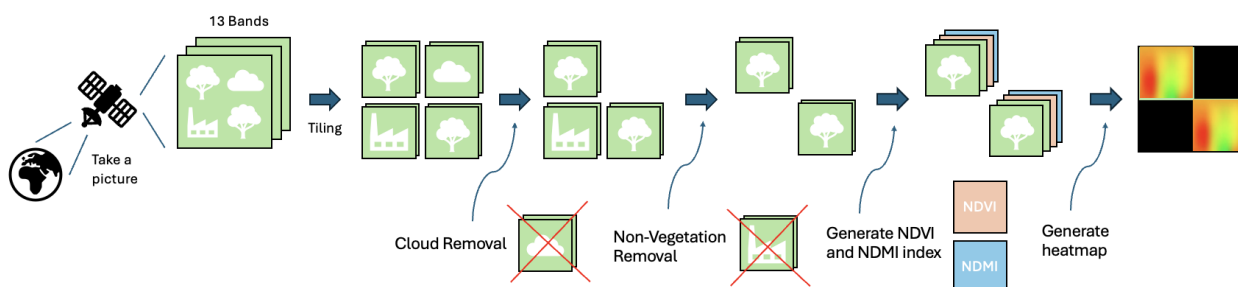


Figure 8: Data flowchart for Fire Prediction Model

5.1 Data Extraction and Ground-Truth Data Generation for Training

The images were downloaded directly from Copernicus Browser with the procedure described in Section 4.3, and all the bands [Fig 9] from Sentinel-2 Level 1C were extracted.



Sentinel-2 Bands	Central Wavelength (μm)	Resolution (m)
Band 1 - Coastal aerosol	0.443	60
Band 2 - Blue	0.490	10
Band 3 - Green	0.560	10
Band 4 - Red	0.665	10
Band 5 - Vegetation Red Edge	0.705	20
Band 6 - Vegetation Red Edge	0.740	20
Band 7 - Vegetation Red Edge	0.783	20
Band 8 - NIR	0.842	10
Band 8A - Vegetation Red Edge	0.865	20
Band 9 - Water vapour	0.945	60
Band 10 - SWIR - Cirrus	1.375	60
Band 11 - SWIR	1.610	20
Band 12 - SWIR	2.190	20

Figure 9: Bands from Sentinel-2 Level 1C MSI, and their corresponding wavelengths and resolutions

Due to their different resolutions (in particular, they spanned from a maximum resolution of 10m, to a minimum of 60m), the bands were resampled to obtain images matching the maximum resolution of 10m of the satellite. The technique used to resample lower-resolution tiles to higher-resolution was to apply a simple 2D nearest-neighbor interpolation of the values [ref. SentinelHub API].

The resulting bands data is put together in a single `numpy.array()`, to allow for easier processing with the following tasks. Information about the bands ordering and the geospatial profile of the image is saved separately. In this way, it is later possible to retrieve the geographical location of the areas for which the model predicts a high risk of fire.

To extract the labels, the processing approach relied mainly on the computation of a pixel value change between images taken before a fire had happened and after it had happened. Specifically, the delta Normalized Burnt Ratio (dNBR) was computed [cf. 1, cf. 2], as

$$dNBR = NBR_{before} - NBR_{after} \quad (3)$$

with

$$NBR = \frac{B08 - B12}{B08 + B12} \quad (4)$$

[ref. for Sentinel-2]

This choice of labeling was motivated by the change in composition of the surface elements after the fire [ref. UNOOSA].

Specifically, the aftermath of a wildfire carries important consequences of the vegetation ecosystem and its reflectance curve, with the most notable variations occurring in the NIR and SWIR ranges. This is caused by a change in the ability of the terrain and vegetation to absorb and reflect water. It is apparent in Figure 11 that the typical "dips" in the SWIR range of the vegetation reflectance curve, which are caused by absorption by water in the plants, disappear in the burnt areas. Also, the high values in the NIR area, associated with the

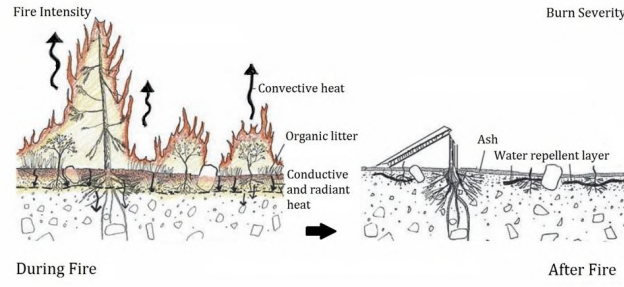


Figure 10: Illustration of fire intensity versus burn severity (Source: U.S. Forest Service)

presence of spongy mesophyll, which is fundamental for photosynthesis, suffer a drastic decrease after the fires.

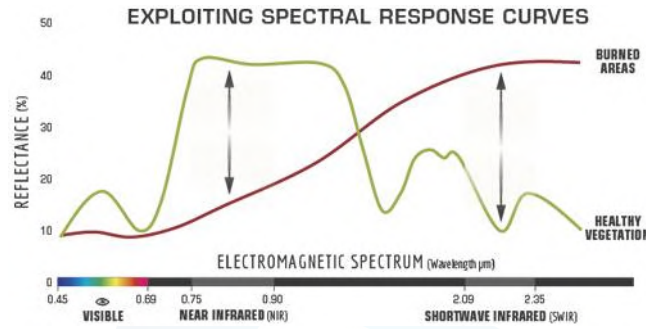


Figure 11: Comparison of the spectral response of healthy vegetation and burned areas. Source: U.S. Forest service.

Following this reasoning, and the fact that other major natural landscapes should change very little, if at all, and in a predictable way, it was possible to associate areas where a fire has occurred with those for which dNBR approaches high values (being the NBR value before largely positive, and the one after largely negative). Values above 0.5-0.6 are typically associated with a moderately severe burn level.

In our specific case, the dNBR between the two closest before and after-images in time to the occurrence of the wildfire was calculated, to extract the label map. Such a map was then normalized into a probability heatmap, with values in the range $[0, 1]$ with a scaled *sigmoid* function, i.e.,

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (5)$$

In our case, the function was reparametrized, with shifting (c) and scaling (k) parameters:

$$\sigma(x) = \frac{1}{1 + e^{-k(x-c)}} \quad (6)$$

Such a scaling was needed due to the specific range of values of dNBR for the various risk classes (e.g., the "moderate risk" middle point corresponded to 0.35, instead of 0), as can be seen in Fig. 12.

After the normalization, to avoid the assignment of high values to pixels corresponding to non-vegetative areas

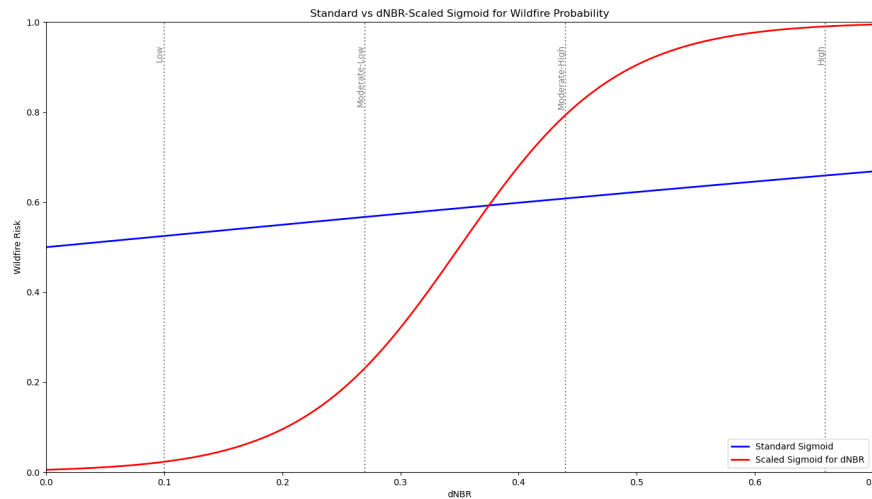


Figure 12: Comparison of the standard sigmoid function (in blue) and the scaled sigmoid used for the probability-mapping of the dNBR labels (in red). Scaling parameters were chosen according to UNOOSA's indications for fire burn severity [13], with probability of 0.5 being assigned to moderate risk by the scaled sigmoid.

Severity Level	dNBR Range (scaled by 10^3)	dNBR Range (not scaled)
Enhanced Regrowth, high (post-fire)	-500 to -251	-0.500 to -0.251
Enhanced Regrowth, low (post-fire)	-250 to -101	-0.250 to -0.101
Unburned	-100 to +99	-0.100 to +0.99
Low Severity	+100 to +269	+0.100 to +0.269
Moderate-low Severity	+270 to +439	+0.270 to +0.439
Moderate-high Severity	+440 to +659	+0.440 to +0.659
High Severity	+660 to +1300	+0.660 to +1.300

Figure 13: UNOOSA's guidelines for fire burn severity.

that underwent modifications between the pre- and post-images (e.g., due to different times of acquisition, clouds, or water bodies natural changes), pixels with an NDVI (Equation 5.4) smaller than 0.2 are set to a value of 0 in the labels. In this way, it is possible to avoid the presence of unwanted "high-risk"-mapped areas in the labels, where they shouldn't be.



Figure 14: Left: the image before a major wildfire occurred in Valparaiso, Chile on February 2, 2024. Center: the image after/during the same fire, on February 7, 2024. Right: the resulting dNBR-based image, with applied NDVI masking, corresponding to the burned area (cf. Copernicus Report)



Finally, before the implementation of the cloud detection and vegetation detection algorithms, the original Sentinel-2 images are divided into tiles of fixed dimension, that is, 256 pixels by 256 pixels. In the filename, information is kept about the relative pixel coordinates of the sub-images inside the larger, full-sized Sentinel-2 image. This allows for the later location retrieval of the images, combining image metadata and tile filenames. The division into smaller composing tiles allows to avoid the downlink of unrelated information, keeping only the relevant non-cloudy, vegetation areas for the fire-risk prediction part.

5.2 Cloud Detection

The world is covered by approximately 65% clouds, which means that in satellite images, around 65% of the information is obscured by cloud cover. Therefore, by implementing an early onboard image filtering system using a cloud detector, we can save approximately 65% of the downlink bandwidth, improving and saving lot of money and time.

For this reason, in this project we have decided to develop an intelligent cloud detection algorithm that operates directly on the satellite platform. This approach allows us to:

- Automatically identify and discard cloud-covered images before further analysis
- Significantly reduce data transmission costs and bandwidth requirements
- Improve overall mission efficiency by prioritizing high-quality data collection

The implementation of this onboard cloud detection system represents a critical optimization that directly addresses one of the most significant challenges in satellite remote sensing operations.

We have evaluated two primary approaches: training a custom model versus utilizing a pre-trained solution. Each option presents distinct advantages and limitations, requiring careful consideration of the trade-offs between accuracy and efficiency.

5.2.1 Pre-trained Solution: s2cloudless Algorithm

We implemented the s2cloudless algorithm, an open-source solution specifically designed for cloud classification in Sentinel-2 imagery. This approach offers several key benefits, as well as some limitations:

Advantages

- High reliability and accuracy in cloud detection
- Well-established and thoroughly tested algorithm

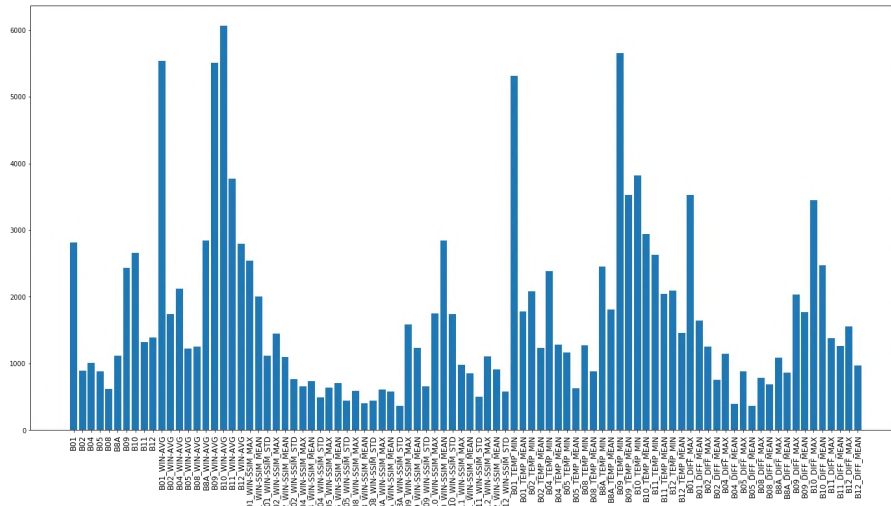


Figure 15: Feature importances for the s2cloudless model

- Proven performance across diverse geographical conditions

Limitations

- Requires 10 spectral bands for processing, significantly increasing computational overhead
- Higher resource consumption due to extensive band requirements
- Reduced processing efficiency compared to lighter alternatives

5.2.2 Custom-trained Solution

Our custom model was developed to address the efficiency limitations of the pre-trained approach while maintaining acceptable performance levels.

Advantages

- Streamlined processing using only RGB + NIR bands (4 bands total)
- Significantly faster execution times
- Improved user experience through reduced processing delays

Limitations

- Lower overall accuracy compared to the s2cloudless algorithm
- Occasional misclassification in challenging scenarios involving snow and rocky terrain
- Reduced robustness due to limited spectral information



5.2.3 Implementation Strategy

Given the complementary strengths of both approaches, we have implemented a dual-solution architecture that allows users to select the most appropriate method based on their specific requirements:

- **Default option:** Custom-trained model (optimized for speed and user experience)
- **High-accuracy option:** s2cloudless algorithm (optimized for precision and reliability)

This flexible implementation enables users to prioritize either processing speed or detection accuracy depending on their particular use case and performance requirements.

5.2.4 Comparison Summary

Metric	Custom Model	s2cloudless
Spectral Bands Required	4 (RGB + NIR)	10
Processing Speed	High	Moderate
Accuracy	Moderate	High
Resource Consumption	Low	High

Table 1: Comparison of cloud detection approaches

5.2.5 s2cloudless Architecture and Implementation

The s2cloudless algorithm implements a pixel-based classification approach utilizing a LightGBM gradient boosting framework trained on a comprehensive dataset of manually annotated Sentinel-2 scenes. The model architecture consists of approximately 170 decision trees with maximum depth constraints ≤ 770 leaves per tree) optimized for computational efficiency while maintaining high classification accuracy.

Each pixel is evaluated independently using the trained ensemble, producing continuous cloud probability estimates between 0 and 1. The pixel-wise approach ensures computational scalability while avoiding artifacts that can arise from spatial smoothing operations in neighboring pixels with heterogeneous surface properties.

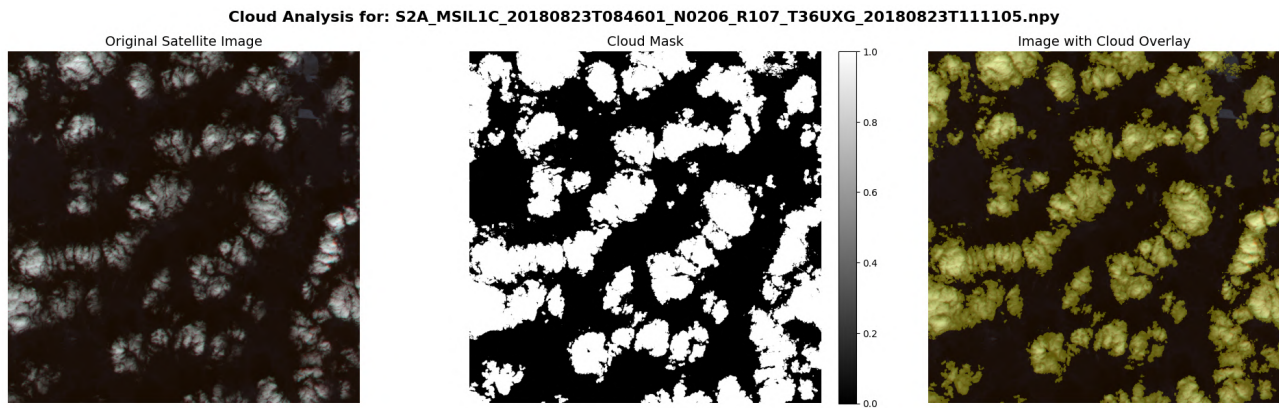


Figure 16: Example of cloud detection using s2cloudless model

5.2.6 Custom-made Model Architecture and Implementation

The cloud detection algorithm is based on a U-Net architecture, which can be seen in sec. 2.3.2. For our purpose, we have built a U-Net that consists of three contracting blocks with 32, 64, and 128 filters, each containing two convolution–batch normalization–ReLU sequences followed by max pooling for downsampling. The decoding path includes three expanding blocks with 64, 32, and the number of output channels filters, each composed of two convolution–batch normalization–ReLU sequences followed by a transposed convolution for upsampling. Feature maps from the contracting path are concatenated into the corresponding expanding blocks, preserving spatial information. During the forward pass, the input is first processed through the contracting path, then progressively upsampled in the expanding path while merging intermediate features at matching resolutions to produce the final output segmentation map.

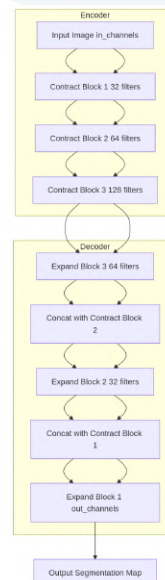


Figure 17: U-Net architecture used for cloud detection

An example of cloud detection using this algorithm is shown below:

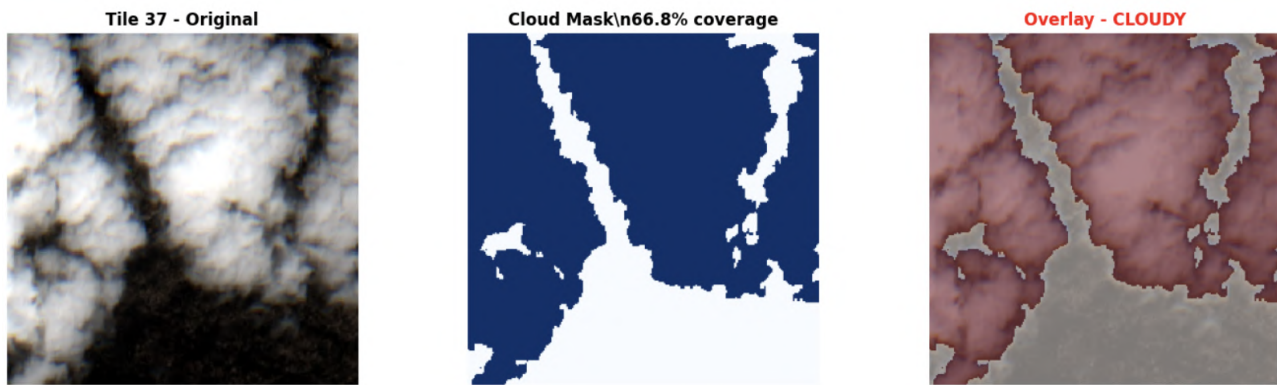


Figure 18: Example result of cloud detection using the U-Net based algorithm

5.3 Vegetation Detection

The vegetation detection algorithm is another necessary part of the fire risk assessment algorithm. The algorithm needs to distinguish landscapes with vegetation and those without. It is needed so that Sentinel-2 only focuses its processing power on certain landscapes, and not overburden itself with processing the risk of fire on landscapes that the user is not interested in. Such landscapes (of no interest to the user) were considered to be those without vegetation (e.g., oceans, deserts, urban areas, etc.).

A Deterministic Algorithm which will directly evaluate the NDVI was created for such purposes. This model extracts the NDVI values of each pixel and compares them to a predetermined threshold (which we set as 0.2) and if the value goes above this threshold the pixel will be labeled as vegetation. Then if a certain percentage of the tile is labeled with vegetation (we have set it as 15%), then the whole tile will be labeled as "vegetation" and it will be kept and passed to the fire risk prediction model.

When checked with 200 images from the Copernicus dataset, **the algorithm achieved an accuracy of 95.5%**



Figure 19: The result of the Deterministic Model: First picture (from left to right) shows the input, second one shows the vegetation level of every pixel, and which tiles have less than 15% of vegetation in them (the ones in beige). The third one shows the output and which tiles were additionally discarded (the ones in red)



5.3.1 Limitations

Even though the vegetation detection algorithm accurately has classified the images, there are some limitations at sight:

1. Complexity - Vegetation patterns are complex, a too simple network may miss subtle differences. This means that it will not be able to classify different biomes, but rather only landscapes with and without vegetation.
2. Channel constraints - The model only uses Red and NIR bands to calculate the NDVI, information from other channels will be lost.

5.4 Feature Extraction

After the vegetation detection step, the remaining tiles are processed to obtain important indices for the later fire-risk prediction task.

In particular, the chosen measures to extract were

- **NDVI:**

$$NDVI = \frac{NIR - R}{NIR + R} = \frac{B08 - B04}{B08 + B04} \quad (7)$$

[ref. for Sentinel-2]

- **NDMI:**

$$NDMI = \frac{NIR - SWIR}{NIR + SWIR} = \frac{B08 - B11}{B08 + B11} \quad (8)$$

[ref. for Sentinel-2]

The *Normalized Difference Vegetation Index* (NDVI) and *Normalized Difference Moisture Index* (NDMI) serve as critical biophysical indicators derived from Sentinel-2 multispectral imagery that provide essential information about vegetation health, density, and moisture content—factors directly relevant to fire risk assessment.

NDVI quantifies vegetation greenness and biomass density by exploiting the distinct spectral properties of healthy vegetation, which strongly reflects near-infrared radiation while absorbing visible red light. This index is normalized, so it ranges from -1 to +1, where values below 0.1 generally indicate non-vegetated surfaces such as water bodies, bare soil, or urban areas, values between 0.1 and 0.3 represent sparse vegetation or stressed plant communities, and values above 0.3 indicate increasingly dense and healthy vegetation cover, with mature forests often exhibiting values between 0.6 and 0.9.



NDMI, conversely, focuses specifically on vegetation water content by leveraging the sensitivity of shortwave infrared bands to leaf moisture, providing crucial information about plant water stress and drought conditions. NDMI values also range from -1 to +1, where higher positive values indicate well-hydrated vegetation with high moisture content, values near zero suggest moderate moisture levels, and negative values typically represent water-stressed vegetation or non-vegetated surfaces.

These spectral indices prove indispensable for fire risk prediction applications because they capture complementary aspects of vegetation condition that directly influence fire susceptibility and behavior. When incorporated

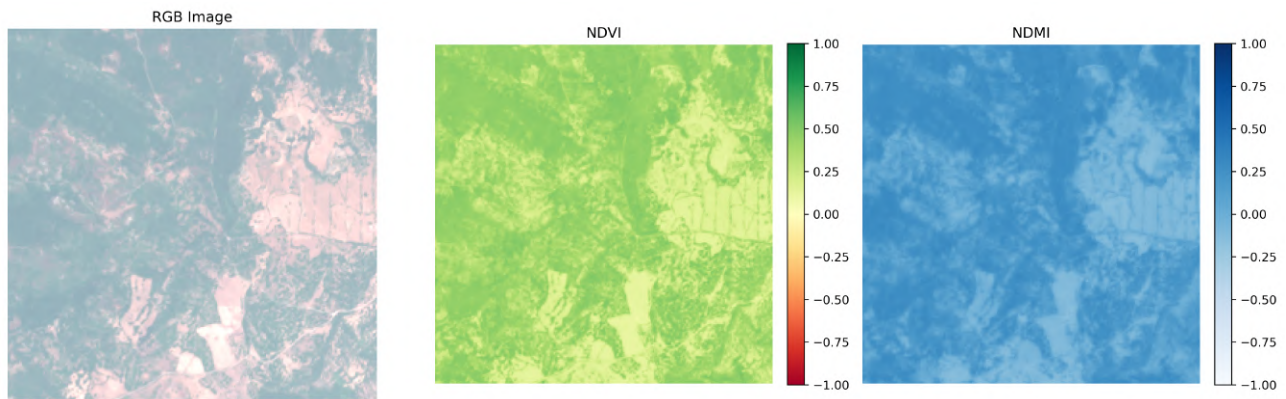


Figure 20: Example of indices for a single tile. Left: RGB Image, Center: NDVI, Right: NDMI

as input dimensions to neural network architectures for fire risk prediction, NDVI and NDMI provide the model with physically meaningful features that capture the fundamental environmental conditions governing fire behavior. The spatial resolution of Sentinel-2 imagery enables the capture of fine-scale heterogeneity in vegetation conditions, allowing models to identify localized areas of elevated risk that might be missed by coarser resolution datasets. The complementary nature of NDVI and NDMI ensures that the neural network receives comprehensive information about both vegetation health and moisture status, enabling more accurate risk stratification and improved prediction performance, especially when combined with bands information compared to models relying on single spectral bands or individual indices alone.

5.4.1 Training Data Augmentation and Normalization

Due to the characteristics of the task, the training dataset tensors consisted mainly of areas with low risk of fire (with 90/95% prevalence). Given this extremely skewed distribution of the dataset, a balancing procedure was needed. This was performed in the following way, passing through all images in the training set:

1. Extract data from a tile label
2. Determine the percentage of pixels with a moderately high risk of fire (> 0.6)
3. If the percentage of pixels with such values is *above* 15%, apply data augmentation, consisting in rotations by 90 and 180 degrees, horizontal and vertical flipping, and addition of Gaussian noise.



4. If the percentage of moderate-risk pixels is *below* 15%, it may be discarded, with 50% probability

The above process is not only useful for rebalancing the dataset, but also for introducing randomness in the characteristics of the training data. In fact, this step can be considered as a regularization step, forcing the model to generalize better to different input types.

In addition to data augmentation, to improve the stability and convergence of the model, a standardization of the data was performed, prior to training. This was done using the `Normalize` class from `torchvision.transforms`, which performed a standardization of the bands, resulting for all bands in a mean of 0 and standard deviation of 1. The statistics of the training dataset were saved in json files, to be later used at the inference phase, before data is passed through the neural network.

5.5 Fire Risk Prediction

The task of fire risk prediction is performed using *Deep Learning* (DL). Specifically, a neural network is implemented to obtain the desired output heatmap of fire risk values, starting from the input image tensor.

More in detail, the input of the network consists of a tensor of dimension (5, 256, 256), where the 5 channels dimensions are given by:

- The NIR band (B08)
- The SWIR1 band (B11)
- The SWIR2 band (B12)
- The computed NDVI
- The computed NDMI

The choices of the bands are motivated by their effectiveness at detecting vegetation moisture, structure, health and water content.

The output of the model, on the other hand, is a 2D matrix, with dimension (256, 256), and values ranging from 0 to 1, to provide a risk heatmap that should match as well as possible the created labels.

For such a regression task, in which we have to produce "probabilities" of fire for each pixel, a U-Net architecture [Sec. 2.3.2] was implemented.



5.5.1 Implementation Details

The specific UNet architecture used was taken from `pytorch`'s own library of models, and in particular, the specific code implementation for the model was obtained from this GitHub repository. As of the time of writing, what was actually used for training involved only minor modifications of the architecture characteristics, to make it suitable to handle the fire prediction data.

The initially chosen loss function was the **Mean Squared Error (MSE)**, commonly used for regression tasks:

$$MSE = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W (I_{h,w} - \hat{I}_{h,w})^2 \quad (9)$$

where $I_{h,w}$ is the target intensity for pixel at position (w, h), and $\hat{I}_{h,w}$ the predicted value.

As optimizer for the weights update, the selected method was Adam [15], which automatically adjusts learning rates for each parameter while maintaining momentum in the loss minimization direction, making it faster and more stable than basic gradient descent.

To evaluate the model's performance, other than the MSE loss itself, **Mean Absolute Error (MAE)** and **Root Mean Squared Error (RMSE)** were chosen as additional measures:

$$\begin{cases} MAE = \frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W |I_{h,w} - \hat{I}_{h,w}| \\ RMSE = \sqrt{\frac{1}{HW} \sum_{h=1}^H \sum_{w=1}^W (I_{h,w} - \hat{I}_{h,w})^2} \end{cases} \quad (10)$$

The model was trained on a 4GB VRAM NVIDIA RTX 500 Ada Generation GPU for 100 epochs, with a batch size of 16 and Adam optimizer, with learning rate of 0.001.

5.6 System Implementation

The system is implemented as a RESTful API using FastAPI with an asynchronous, job-based processing model to handle computationally intensive satellite image analysis tasks. The architecture is organized into a web API layer, a background processing engine, and modular utilities responsible for each analysis stage. FastAPI provides the HTTP interface with automatic OpenAPI documentation, while Pydantic models ensure request/response validation and serialization. CORS middleware is configured to enable seamless frontend integration.

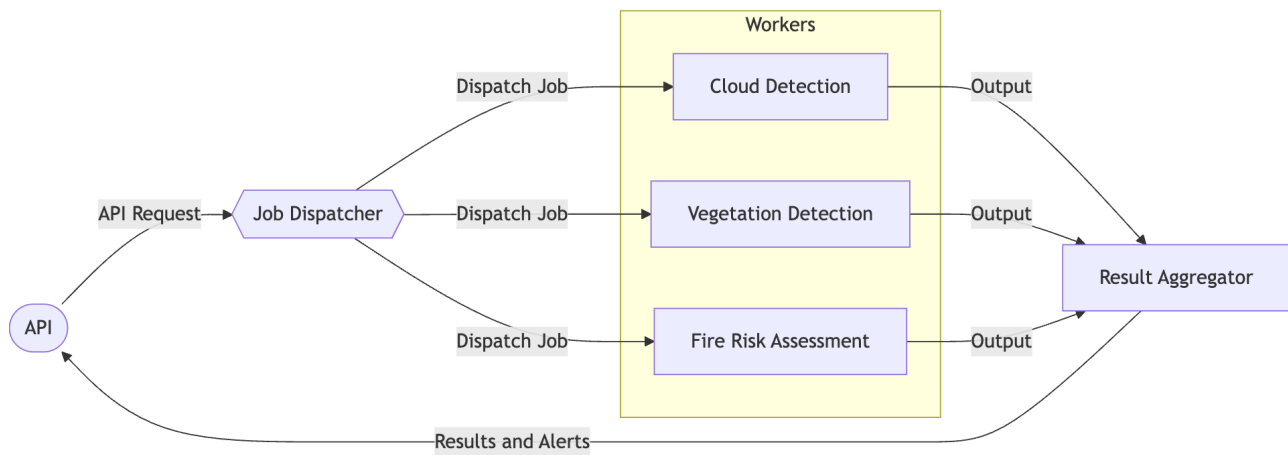


Figure 21: System implementation showing API-driven processing workflow.

Image Processing Pipeline

Input images are provided as NumPy arrays (.npy) containing 13 spectral bands. The system validates band count, cropping excess bands when necessary, and divides large scenes into 256×256 pixel tiles for parallel, memory-efficient computation. Processing follows the sequential pipeline described before:

1. Cloud detection.
2. Forest detection on cloud-free tiles.
3. Fire risk prediction using vegetation and moisture indices.

Asynchronous Job Management

Processing is managed using FastAPI background tasks. Each request generates a unique UUID job identifier for progress tracking. Job states include *pending*, *processing*, *completed*, and *failed*, with detailed metrics such as tile counts, success rates, and real-time completion percentages.

Cloud Detection

Cloud detection processes individual tiles using threshold-based spectral analysis combined with machine learning classification. Each tile produces a binary mask and cloud coverage percentage. Visual outputs include:

- RGB composites with blue overlays for clouds.
- Binary cloud masks for downstream analysis.



Forest Detection

Forest detection is restricted to cloud-free tiles. The system computes the Normalized Difference Vegetation Index (NDVI) from red and near-infrared bands to classify vegetation cover. Outputs include per-tile binary masks, vegetation percentage scores, and composite maps with green overlays highlighting forested areas.

Fire Risk Prediction

The fire risk module calculates both NDVI and the Normalized Difference Moisture Index (NDMI) using NIR, red, and SWIR bands. Machine learning models generate fire probability maps, which are visualized as heatmaps ranging from low-risk (blue/green) to high-risk (red).

Error Handling

Robust error handling provides clear diagnostics and placeholder outputs when failures occur. Background tasks are isolated so that errors in one job do not affect concurrent processing. The modular design enables independent testing and maintenance of each component.

6 Experimental Results and Evaluation

6.1 Experimental Results of Cloud Detection Model

6.1.1 s2cloudless Algorithm

S2Cloudless is a widely adopted model in the field of Earth observation for cloud detection in Sentinel-2 imagery. Its architecture is specifically optimized for Sentinel-2 data, making it an excellent fit for this type of satellite imagery. Numerous studies have demonstrated its superior performance compared to alternative approaches.



		Fraction of classifications as clouds			
		MAJA	Fmask	Sen2Cor	Sentinel Hub 
True Label	Cloud	99,4%	95,8%	95,9%	100%
	Cirrus	64,1%	89,8%	95,1%	73,7%
	Land	0,7%	12,0%	9,6%	7,0%
	Water	0,0%	0,0%	0,0%	0,0%
	Snow	10,6%	71,5%	58,0%	18,3%
	Shadow	5,8%	8,1%	8,4%	3,2%

Figure 22: Cloud and cirrus cloud detection rates and land, water, snow and shadow misclassification rates as clouds as determined using 15 Sentinel-2 scenes hand labeled by Hollstein et al.

In this work, we evaluated the effectiveness of the S2Cloudless model using a dedicated test dataset. The dataset consisted of Sentinel-2 images annotated with difficulty levels for cloud detection on a scale from 1 (easy) to 5 (very challenging).

The model was benchmarked against these varying complexity levels. The results were promising: while performance decreased slightly for the most challenging cases, the model consistently achieved high accuracy across the majority of scenarios. The next figure illustrated the performance distribution across difficulty levels.

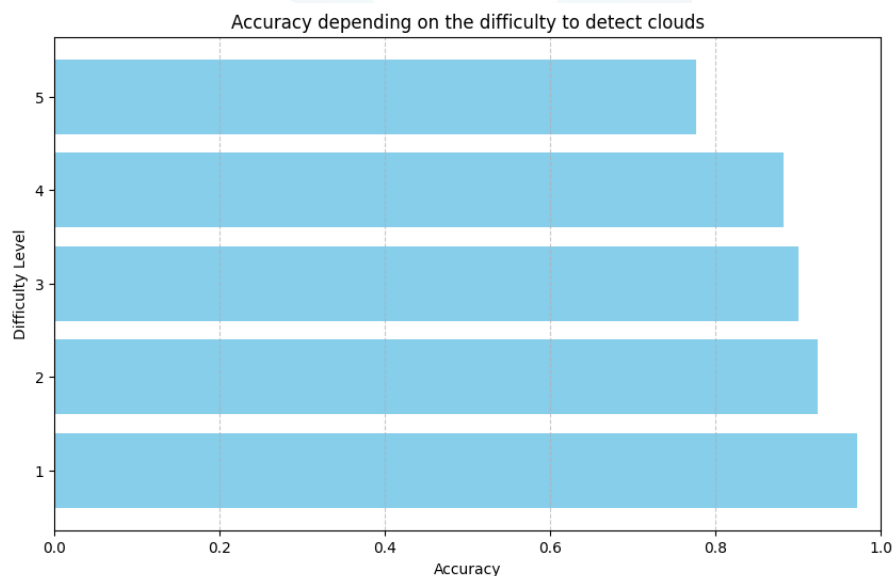


Figure 23: Accuracy of s2cloudless depending on the difficulty to predict clouds depending on the images



6.1.2 Custom-made Algorithm

The home-made cloud detection model shows solid performance across all evaluation metrics, although it performs slightly below the widely used S2cloudless model:

- **Accuracy:** - The model correctly classifies approximately 87% of all pixels in the dataset.
- **Precision:** - When predicting a pixel as cloud, the model is correct about 89% of the time.
- **Recall:** - The model successfully identifies roughly 82% of all actual cloud pixels.
- **F1-Score:** - The harmonic mean provides a balanced measure between precision and recall.

Overall, while slightly inferior to S2cloudless, the model behaves reliably and can be considered a valid approach for cloud detection in Sentinel-2 imagery.

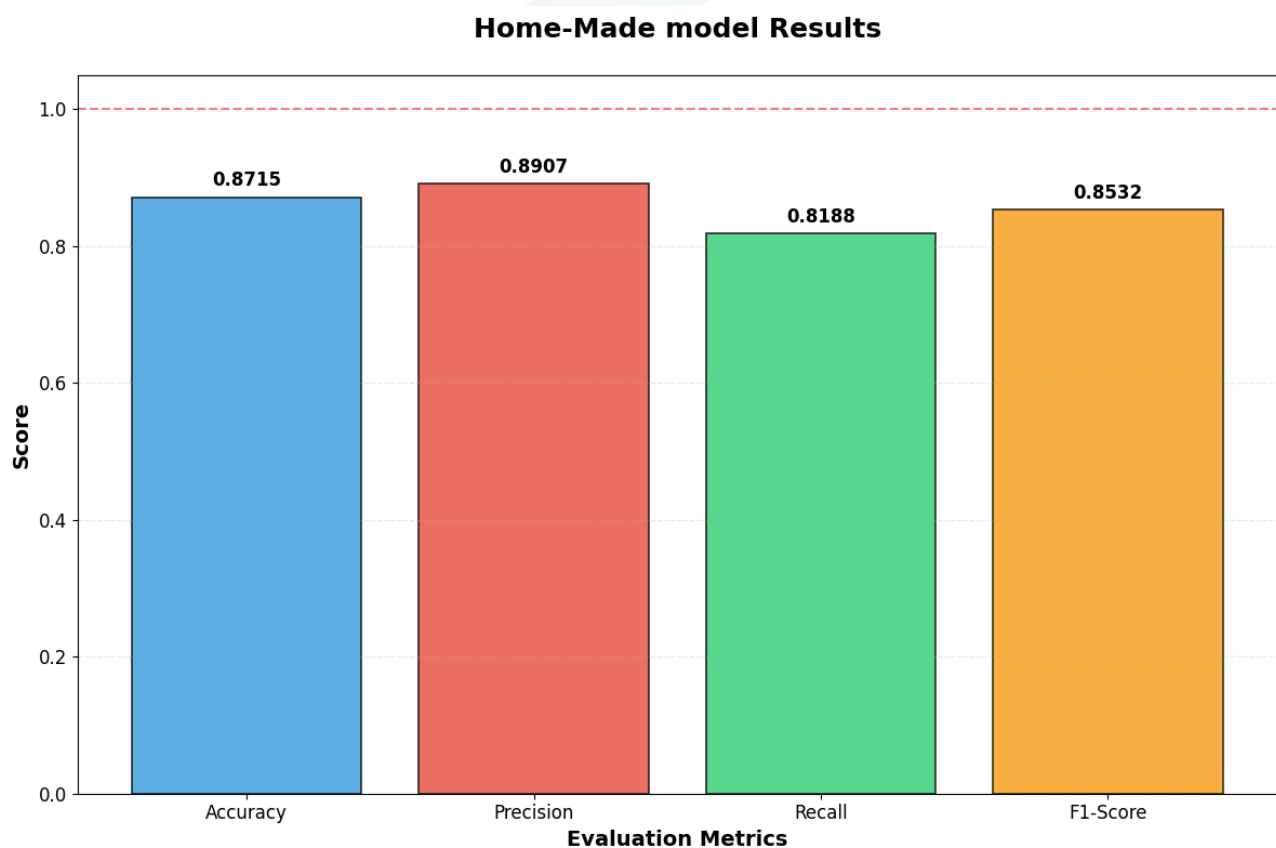


Figure 24: Performance comparison of the custom-made cloud detection model.



6.2 Experimental Results of Fire Prediction Model Training

The fire prediction model managed to reach a low MAE on the train set, around 5% for the final epoch, and comparable (although not as good) results on the validation set, suggesting the balancing step in the preprocessing pipeline helped with the previous problem of producing the same, low-risk output. However, the training procedure adopted showed a certain degree of overfitting to the training set, especially when the performances were evaluated against the external test set of 1800+ tile extracted from a full-sized Sentinel-2 image from Greece (see Table 2). This suggests that the model possibly needs possible regularization procedures, like the addition of dropout layers in the architecture, or the reduction of its complexity. Specifically, such prototype training run did not give enough importance to such areas, given their very low representativeness of this initial testing, unprocessed dataset, so it achieved a really good performance in mapping the areas at a medium risk score, but performed quite poorly on the target fire areas (see Figs. 25, 26).

A further and more tailored evaluation procedure could involve the analysis of the model's ability to correctly individuate tiles corresponding to regions with areas of high fire risk, even if the error between the ground truth label and the model produced heatmap is not null. This would allow to consider more practically the real-world impact of such a prediction model, with the implementation of the model being associated with the precise geolocalization of the single-tile area. Associating an overall "class of risk" with an area, and connecting to it the corresponding real-world coordinates could prove extremely impactful on the actual prevention of wildfire risk

Dataset	MAE	RMSE
Train (best)	0.0501	0.1297
Validation (best)	0.0766	0.2002
Test	0.0911	0.2190

Table 2: MAE and RMSE performances of the model after a training process of 100 epochs. The training set was composed of tiles taken from four large images, representing four different areas, and the train/val dataset split was of 80/20. The model was tested on other 1800+ tiles from a different image.

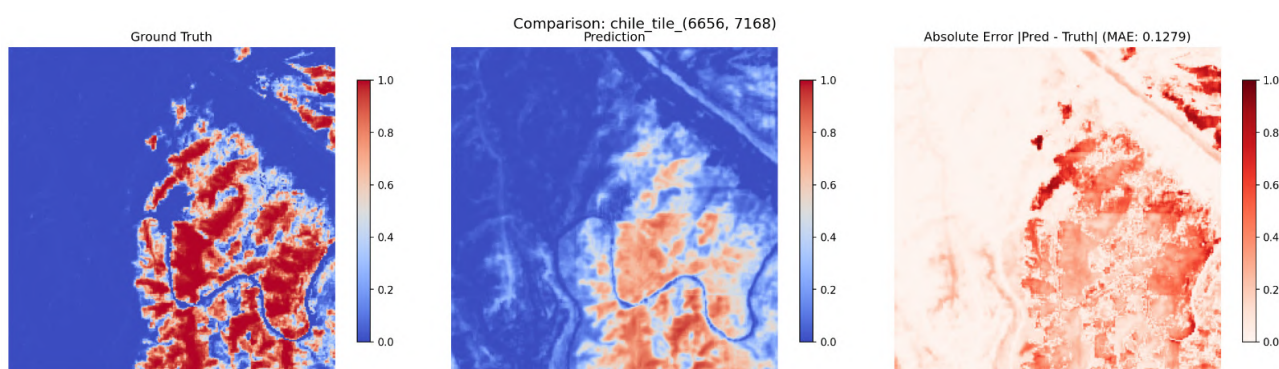


Figure 25: Comparison of the ground truth label (left), model's output (center) and error between the two (right) on a Chile training tile.

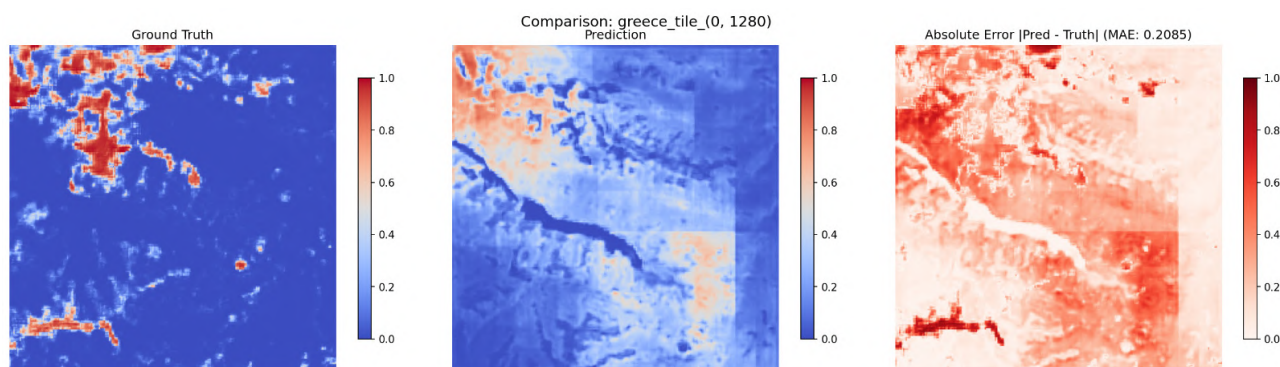
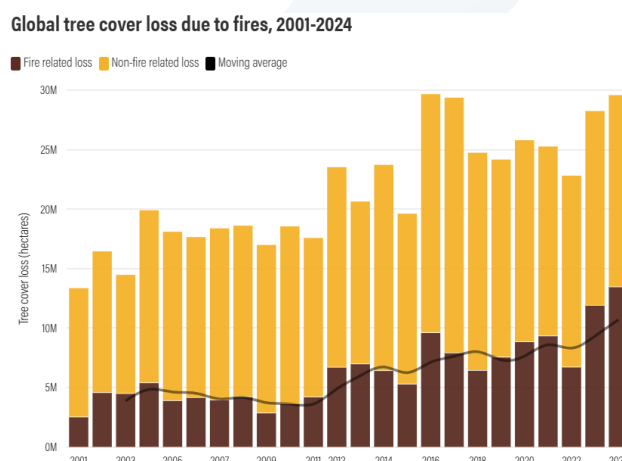


Figure 26: Comparison of the ground truth label (left), model's output (center) and error between the two (right, on a Greece test tile).

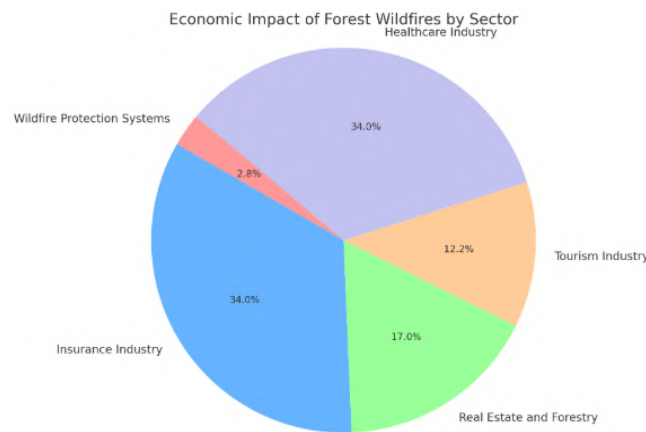
7 Business and Societal Impact

7.1 Business Impact

Forest fires inflict significant direct and indirect losses on businesses. Direct losses include substantial property damage to businesses and infrastructure. Fires also cause business interruption, halting operations and leading to lost revenue. As fire risks escalate, businesses face increased insurance costs and may even struggle to secure coverage. Furthermore, the timber industry suffers considerable losses from the destruction of valuable forest resources, with climate change impacts on forestry projected to lead to 4 billion dollars in export revenue losses for Canada alone.[16, 17, 18]



In addition to these direct impacts, wildfires cause extensive indirect business losses. Supply chain disruptions are common as fires block transportation routes and limit raw material availability, causing delays and higher expenses. The tourism sector is particularly vulnerable, as wildfires reduce tourism and negatively affect related businesses. Property values can also decline in affected regions, impacting businesses with real estate holdings. Wildfire smoke leads to higher health costs from associated respiratory and cardiovascular issues, and the overall economic disruption can cause a significant decline in regional GDP growth, with some events, like the Los Angeles wildfires, having an economic impact exceeding **250 billion dollars**. [19, 20]



We are developing a wildfire prediction module that will be more than a scientific tool. By providing foresight into wildfire risks, our technology will deliver measurable economic benefits:

- For Insurance Companies: Our module will enable more accurate risk assessment models, helping to price property and agricultural coverage with greater precision. It will also facilitate new products like parametric insurance, which provides fast payouts to businesses for a rapid recovery.[21, 22]
- For Banks and Financial Institutions: The module will be a critical tool for managing credit risk. By integrating our predictive data, banks can refine investment strategies, build more resilient portfolios, and safeguard assets from potential losses.
- For Governments: Our module will empower governments with the data needed for intelligent land-use planning and stricter building codes. This proactive approach will reduce fire risks, minimize suppression costs, and protect infrastructure. It will also help them transition from reactive firefighting to proactive prevention.[23, 24, 25]

7.2 Societal Impact

Wildfires pose a significant threat to communities and ecosystems, impacting everything from human health to cultural heritage. Wildfires destroy biodiversity and natural habitats. Remote communities are particularly vulnerable, suffering from frequent evacuations and the loss of material, economic, and social resources. Many remote and Indigenous communities face limited access to resources and fragile infrastructure, such as power lines and communication cables, which can be easily destroyed. Wildfires also have devastating impacts on cultural monuments, which are often irreplaceable. The physical damage from fire and smoke can destroy or weaken structures.

Wildfires have significant consequences for both physical and mental health:

- Physical Health: Smoke from wildfires causes respiratory problems, such as asthma and bronchitis, as



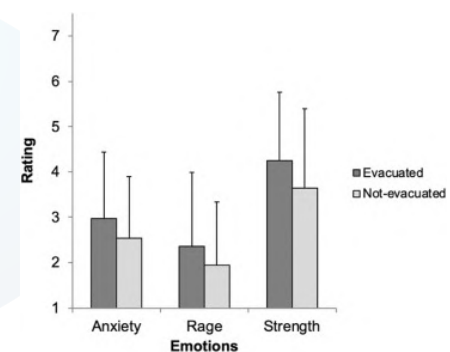
Figure 27: The consequences of the forest wildfire. Before and after: South Korea's wildfires turn temple to ash

well as an increased risk of cardiovascular disease. Children, the elderly, and those with chronic health conditions are especially vulnerable.

- **Mental Health:** The psychological trauma of fires and evacuations can lead to depression and PTSD, especially in children and adolescents. Indigenous communities, who are frequently exposed to evacuations, report significant stress from the disruption of their social structures.

Year	Number of fire deaths	Population	Fire death rate (per million population)
2014	3,428	318,386,329	10.8
2015	3,362	320,738,994	10.5
2016	3,515	323,071,755	10.9
2017	3,645	325,122,128	11.2
2018	3,810	326,838,199	11.7
2019	3,515	328,329,953	10.7
2020	3,790	331,526,933	11.4
2021	4,316	332,048,977	13.0
2022	4,446	333,271,411	13.3
2023	4,371	334,914,895	13.1
10-year trend (%)			+27.0%

(a) The fire deaths (2014 - 2023)



(b) Mentality of the evacuated and non-evacuated

Our wildfire prediction module will transform wildfire management from a reactive to a proactive approach. By analyzing vast data sets our system will identify high-risk areas with unparalleled accuracy:

- **Provide Early Warnings:** This capability will provide communities and local authorities with crucial early warnings, giving them time to prepare. For example, municipalities could receive alerts hours or days before a fire is likely to ignite.
- **Optimize Evacuations:** The data will allow for more effective, targeted evacuations, minimizing panic and ensuring the safety of those most at risk.
- **Empower Firefighting Agencies:** The module will be a powerful asset for agencies like the U.S. Forest Service or Cal Fire. It will help them orchestrate optimized firefighting efforts by strategically deploying personnel, aircraft, and equipment to where they are most needed.



8 Team and Work Breakdown

8.1 Team



Dario Carretta

Designs the data pipeline and develops the fire prediction model that powers the team's core solution.



Sercan Kahvecioğlu

Captains the team by driving project planning, task distribution and dockerization while pitching the business vision with clarity and impact.



Diego Luque

Drives the development of the cloud detection model and architects the execution pipeline that ensures the system runs seamlessly.



Lior Mayer

Develops the vegetation model and crafts compelling presentations that showcase the team's innovation.



Ivana Moskova

Leads research and data acquisition, providing critical insights that strengthen the team's models and strategy.

Figure 29: Team



8.2 Work Breakdown

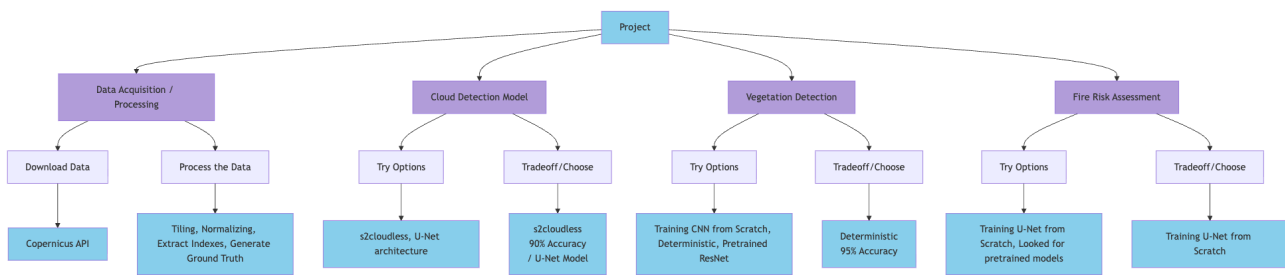


Figure 30: Project overview

9 Conclusion

This document illustrates the capacity of the Forest Wildfire Risk Assessment Module to integrate artificial intelligence (AI)-driven analytics and high-resolution satellite imagery, thereby effecting a paradigm shift from reactive firefighting to proactive, with precision-based wildfire risk monitoring. This shift enables more expeditious and efficient intervention.

Throughout this document, we have detailed the development of this module, leveraging an AI algorithm, Sentinel-2 imagery analysis, robust cloud and forest detection techniques, advanced feature extraction, and the U-Net model for precise fire risk assessment. The provision of early warnings with high spatial resolution will empower decision-makers with actionable information, thereby facilitating efficient and targeted evacuations, optimized resource allocation, and data-driven policy enforcement. Beyond the operational efficiencies it engenders, this technology has the potential to serve as a catalyst for fostering more resilient communities, safeguarding economic stability, and preserving invaluable natural ecosystems.

The solution's architecture is modular and scalable, which ensures its adaptability across diverse geographical regions, forest biomes, and climatic conditions. This lays the groundwork for a **global wildfire risk monitoring infrastructure**. However, it is important to note that this advancement is not a panacea. In order to effectively address the challenges posed by evolving climate dynamics and data integration challenges, there is a necessity for sustained interdisciplinary collaboration, forward-looking policy frameworks, and ongoing investment in AI research.

Subsequent research endeavors will concentrate on enhancing the precision of predictions in extreme environmental circumstances, broadening the scope of multi-sensor data fusion capabilities, and incorporating socio-economic risk layers to provide a comprehensive situational awareness platform. Collaboration with international fire management agencies will be pivotal in translating technological innovation into operational effectiveness on a global scale.



In a world increasingly confronted with the realities of climate change, innovative solutions such as the Forest Wildfire Risk Assessment Module are imperative. We stand at the forefront of transforming wildfire management, moving from a reactive stance to a proactive watchfulness, and ultimately fostering a safer, more resilient planet for all.

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